

Exam 1 Equation Sheet

(Additions and minor changes may be made by the professor through Thursday, 9/24.)

Absolute: E
Relative (percent): ε
Approximate: a
True: t
Specified: s

```
function GAUSS_PIVOT([A], {b})
  for k ← 1 to n - 1 do
    p ← argmaxi ≥ k |aik|
    swap rows k and p of [A] and {b}
    for i ← k + 1 to n do
      f ← aik/akk
      for j ← k to n do
        aij ← aij - f akj
      bi ← bi - f bk
    xn ← bn/ann
    for i ← n - 1 down to 1 do
      xi ← (bi - ∑j=i+1n aij xj)/aii
  return {x}
```

$$\sum_{i=k}^m a f(i) = a \sum_{i=k}^m f(i)$$

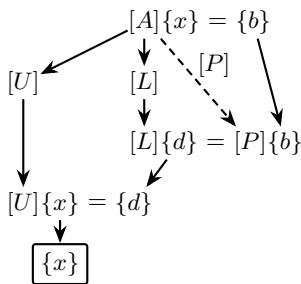
$$\sum_{i=k}^m [f(i) + g(i)] = \sum_{i=k}^m f(i) + \sum_{i=k}^m g(i)$$

$$\sum_{i=1}^m 1 = m \quad \sum_{i=k}^m 1 = m - k + 1$$

$$\sum_{i=1}^m i = \frac{m(m+1)}{2}$$

$$\sum_{i=1}^m i^2 = \frac{m(m+1)(2m+1)}{6}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = [P][A]$$



$$\text{cond}[A] = \|[A]\| \|[A]^{-1}\|$$

$$\frac{\|\{\Delta x\}\|}{\|\{x\}\|} \leq \text{cond}[A] \frac{\|\{\Delta A\}\|}{\|[A]\|}$$

$$\frac{\|\{\Delta x\}\|}{\|\{x\}\|} \leq \text{cond}[A] \varepsilon$$

$$2.22 \times 10^{-16} \quad 1.19 \times 10^{-7}$$

$$E_a = \frac{\Delta x}{n-1} \quad E_a = \frac{\Delta x}{2^n}$$

$$n = \frac{\Delta x}{E_s} + 1 \quad n = \log_2 \left(\frac{\Delta x}{E_s} \right)$$

```
function NEWTON(f, f', x0)
  x ← x0
  for k ← 1 to maxit do
    Δx ← -f(x)/f'(x)
    x ← x + Δx
    if |Δx| < atol then return x
  report failure
```

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

$$\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100\%$$

$$[J](\{x_i\}) \{\Delta x\} = -\{f\}(\{x_i\})$$

$$\{x_{i+1}\} = \{x_i\} + \{\Delta x\}$$

$$E_{a,i+1} = \frac{E_{a,i}}{2} \quad E_{t,i+1} \approx \frac{-f''(x_r)}{2f'(x_r)} E_{t,i}^2$$

$$[J] = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

$$\text{Forward: } f'(x) \approx \frac{f(x+h) - f(x)}{h} \quad h \approx \sqrt{\varepsilon} x$$

$$\text{Central: } f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

$$\frac{\partial \{f\}}{\partial x_j} = \frac{\{f\}(\{x\} + h\{e_j\}) - \{f\}(\{x\})}{h}$$

No unique $[A]\{x\} = \{b\}$ solution $\iff \det[A] = 0 \iff [A]$ is singular $\iff [A]$ not full rank $\iff [A^{-1}]$ does not exist.

$$cf(n) = O(f(n)) \quad \forall c > 0$$

$$\text{If } f_1(n) = O(n^a) \text{ and } f_2(n) = O(n^b), \text{ then } f_1(n) + f_2(n) = O(n^{\max(a,b)})$$

$$\text{If } f_1(n) = O(n^a) \text{ and } f_2(n) = O(n^b), \text{ then } f_1(n) \cdot f_2(n) = O(n^{a+b})$$

$$\text{If } |f(n)| \text{ is bounded, } f(n) = O(1)$$

abs(x) $|x|$
sign(x) $-1, 0, \text{ or } 1$
norm(x) $\|x\|_2$ (2-norm of vector or matrix)
sqrt(x), exp(x), log2(x)
[m, i] = max(abs([4, -6, 3])) value & index: **m=6, i=2**
n = length(v); [m, n] = size(A)
zeros(m, n), ones(m, n), eye(n)

A(i, :), A(:, j), A(i, j:n)
A' transpose; **A*x; x = A\b**
inv(A), det(A), cond(A)
eps, Inf, NaN, isnan(x), isinf(x)
@(x) x^2 - 3 defines an anonymous function that computes $x^2 - 3$
@sin creates a handle for the function **sin**
&&, ||, ~=, ==, ~